

Assignment 2- Case Study (8995-G level Task)

Stage 5 Lab Activities

Refactoring SmartCare into a Layered Architecture

HUMAN ANALYSIS -> REFACTOR -> AI REVIEW -> VERIFY | 1 hour

A - Inspect SmartCare v0.4

Identify domain, workflow, data-access and presentation responsibilities currently mixed together.

B - Propose Architecture

Draw Presentation -> Service -> Domain, with Service using a Repository abstraction and Persistence implementing it.

C - Create Package Structure

Create domain/, services/, repositories/, persistence/ and presentation/ or a justified equivalent.

D - Introduce AppointmentService

Move workflow coordination into a focused service without stealing Appointment domain behaviour.

E - Repository Abstraction

Define a small AppointmentRepository contract using only current use-case needs.

F - AI Architecture Review

Ask AI to review dependency direction, misplaced responsibilities and unnecessary complexity; request simplest justified improvements.

G - Refactor and Verify

Apply only justified changes and confirm required behaviour remains unchanged.

H - Reflection

Document one AI suggestion accepted, one modified and one rejected/deferred.

Suggested AI prompt

Act as a software architecture reviewer. Review this small SmartCare Python system against separation of concerns, cohesion, coupling and introductory SOLID principles. Identify concrete layer violations and dependency risks. Prefer the simplest refactoring that solves an observed problem. Do not introduce frameworks, microservices or patterns unless current requirements justify them.